THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Three coins are tossed. The probability of getting not all the same is: 1

(a) $\frac{3}{4}$ **(b)** $\frac{1}{2}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{5}$

2. A bag has 10 red balls and some blue balls. If the probability of drawing a blue ball is $\frac{3}{5}$, the number of blue balls is: 1

(a) 16 **(b)** 14 **(c)** 15 **(d)** 17

3. A bag has 45 balls, red or blue. If $P(\text{red}) = \frac{2}{5}$, the number of blue balls is: 1

(a) 28 **(b)** 29 **(c)** 27 **(d)** 26

4. A letter is chosen from the word MATHEMATICS. The probability that it is a vowel or the letter T is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{6}{11}$ **(c)** $\frac{7}{11}$ **(d)** $\frac{5}{11}$

5. The probability that a seed germinates is $\frac{4}{5}$. Out of 3500 seeds the expected number that will NOT germinate is: 1

(a) 699 **(b)** 700 **(c)** 701 **(d)** 702

6. A bag contains only red and blue balls. If $P(\text{red}) = 2P(\text{blue})$, then $P(\text{blue}) =$ 1

(a) $\frac{2}{3}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{4}$

7. Two dice are rolled together. The probability that the product of the numbers is 4 is: 1

(a) $\frac{11}{12}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{13}$ **(d)** $\frac{1}{12}$

8. Two dice are rolled together. The probability that the larger number is exactly 4 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{29}{36}$ **(c)** $\frac{7}{37}$ **(d)** $\frac{7}{36}$

9. **Assertion (A):** For any event E, $P(\text{not } E) = 1 - P(E)$. 1

Reason (R): For any event E, $P(E) + P(\text{not } E) = 2$.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The probability of an event can be 1.5. 1

Reason (R): The probability of a sure event is 1.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A bag has 15 red balls and some blue balls. If $P(\text{blue}) = \frac{2}{5}$, find the number of blue balls. 2

12. Find the probability that the larger of the two numbers on rolled dice is exactly 3. 2

13. Two dice are rolled together. Find the probability that the product of the numbers is 8. 2

14. A bag has 15 red balls and some blue balls. If $P(\text{blue}) = \frac{3}{8}$, find the number of blue balls. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains red, blue and green balls. $P(\text{red}) = \frac{2}{8}$ and $P(\text{blue}) = \frac{3}{8}$ and there are 48 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

16. In 150 throws of a die, the number of times a prime number appeared was 80 and a number greater than 4 appeared 46 times (these include 22 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

17. A letter is chosen at random from the word EXAMINATION. Find the probability that it is 3
(i) not N,
(ii) a vowel or N.

OR

A bag contains red, blue and green balls. $P(\text{red}) = \frac{1}{6}$ and $P(\text{blue}) = \frac{3}{6}$ and there are 36 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

18. In 120 throws of a die, the number of times a prime number appeared was 66 and a number greater than 4 appeared 38 times (these include 18 throws showing 5). Find the experimental probability of 3
(i) a prime number,
(ii) a number greater than 4,
(iii) a prime number or a number greater than 4.

OR

A letter is chosen at random from the word PROBABILITY. Find the probability that it is 3
(i) not B,
(ii) a vowel or B.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A shopkeeper in Pune notes how 200 customers paid one day: 90 paid by cash, 70 by UPI and 40 by card. A customer is chosen at random from these 200. 5
(i) Find the experimental probability that the customer paid by cash.
(ii) Find the probability that the customer paid by UPI.
(iii) Find the probability that the customer paid by card.
(iv) Find the probability that the customer did not pay by cash, and estimate how many of the next 500 customers will use UPI.

20. Two dice are rolled together. Find the probability that 5
(i) the sum is 9,
(ii) the sum is more than 9,
(iii) the sum is at most 9. Verify that
(ii) and
(iii) add to 1.

OR

A bag has 6 red, 8 blue and 12 green balls. Some green balls are removed so that $P(\text{green})$ becomes $\frac{1}{3}$. How many green balls were removed? 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND